

Areta 360 Assignment - Solution

AI-Powered Schema Selection for NL to SQL Pipelines

I have created a simple project using Python and Django framework. It allows users to convert natural language queries to provide relevant schema and relevant Sql queries. I have exposed one endpoint.

I have attached .zip file for my python code which has Django project. I'm attaching screenshot at end of this documentation to showcase that I was able to run it locally.

For example:

- **Input:** "Get all users who registered in the last 30 days"
- **Output:**
 - It provides Users schema
 - "SELECT * FROM users WHERE created_at >= NOW() - INTERVAL 30 DAY;"

How It Works

1. A user sends a **natural language query** to the API.
2. The system **identifies relevant tables** from the database.
3. The query is **converted into SQL** using AI (GPT-3.5-Turbo).
4. The **SQL query is returned** as a response along with relevant models.

API Endpoint

Method	URL	Description
POST	/api/nl-to-sql/	Convert a natural language query into an SQL query

Request Format

POST request

```
{  
  "query": "Show me all employees who joined in 2024"  
}
```

Response Format

Api response:

```
{  
  "natural_query": "Show me all employees who joined in 2024",  
  "relevant_tables": [  
    {  
      "name": "employees",  
      "columns": ["id", "name", "email", "hire_date"]  
    },  
    ],  
  "sql_query": "SELECT * FROM employees WHERE hire_date >= '2024-01-01';"  
}
```

Project Structure

```
assignment/  
|-- urls.py      # API routes  
|-- views.py     # API logic to process NL queries  
|-- services.py  # Handles table selection & SQL generation  
|-- schema_loader.py # Loads database schema  
|-- schema.json  # Stores database tables & columns
```

Key Components

1 QueryProcessor (Finds Relevant Tables)

- Extracts **important words** from the query.

- Matches the **best-fit database tables** using python libraries.

2 SQLGenerator (Creates SQL Query)

- Uses **GPT-3.5-Turbo** to generate an SQL query.
- Takes **query + relevant tables** as input.

Setup & Run the Project

1 Install Dependencies

```
pip install django openai spacy sentence-transformers  
python -m spacy download en_core_web_sm  
python manage.py runserver
```

Screenshots -

ActivitiesPostman

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Postman

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Working locally in Scratch Pad. Switch to a Workspace

POST http://loGET http://locPOST http://loGET http://locGET http://locGET http://locPOST http://loGET http://locPOST http://loGET http://locPOST http://loGET http://locPOST http://loGET http://locPOST http://loGET http://locPOST http://lo

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No Environment

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⌵

http://localhost:8000/api/nl-to-sql/

Save

⌵

⌵

POST http://localhost:8000/api/nl-to-sql/

Send

ParamsAuthorizationHeaders (9)BodyPre-request ScriptTestsSettings

noneform-datax-www-form-urlencodedrawbinaryGraphQLJSON

1 {
2 "query": "Show me all orders placed last week."
3 }
4

BodyCookiesHeaders (8)Test Results

Status: 200 OKTime: 3.66 sSize: 591 BSave Response

PrettyRawPreviewVisualizeJSON

1 {
2 "natural_query": "Show me all orders placed last week.",
3 "relevant_tables": [
4 {
5 "name": "orders",
6 "columns": [
7 "order_id INTEGER PRIMARY KEY",
8 "customer_id INTEGER",
9 "order_date DATETIME",
10 "status VARCHAR(20)",
11 "total_amount DECIMAL(10,2)",
12 "payment_method VARCHAR(50)",
13 "shipping_address TEXT"
14]
15 }
16],
17 "sql_query": null
18 }

Find and Replace

Preview

Runbook

Trash

```
test----->
❌ OpenAI API Error: Error code: 429 - {'error': {'message': 'You exceeded your current quota, please check your plan and billing details. For more information on this error, read the docs: https://platform.openai.com/docs/guides/error-codes/api-errors.', 'type': 'insufficient_quota', 'param': None, 'code': 'insufficient_quota'}}
[28/Feb/2025 16:10:10] "POST /api/nl-to-sql/ HTTP/1.1" 200 315
/home/kk/Arata360/assignment/assignment/services.py changed, reloading.
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).
February 28, 2025 - 16:13:10
Django version 5.1.6, using settings 'assignment.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CONTROL-C.
```